

Corrosion of Thermo-Mechanically Treated (TMT) Mild Steel in Reinforced Concrete Structures: Mechanical and Structural Implications

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Abstract

Reinforced concrete (RC) remains the most widely used construction material due to its combined compressive and tensile strength principally due to the use of thermo-mechanically treated (TMT) mild steel for the reinforcement. However, the long-term performance of RC structures is severely threatened by corrosion of embedded mild steel reinforcement, particularly in chloride-rich and carbonated environments. Corrosion undermines the protective alkalinity of concrete, initiating electrochemical reactions that generate expansive rust products, causing cracking, delamination, and spalling of cover concrete. Mechanically, corrosion reduces the cross-sectional area, ductility, and bond strength of reinforcement, while structurally it leads to diminished load-bearing capacity, serviceability issues, and in extreme cases, catastrophic collapse. Evidence from global case studies of bridges, parking garages, marine piers, and residential buildings illustrates the socio-economic cost of corrosion-related deterioration. This study synthesizes existing knowledge on corrosion mechanisms, their mechanical and structural

implications, and highlights contemporary mitigation approaches. Strategies such as high-performance concretes, corrosion-resistant reinforcement, cathodic protection, coatings, and advanced monitoring technologies are discussed as means to enhance durability. The findings underscore the urgent need for durability-based design, proactive maintenance, and adoption of advanced materials, especially in developing regions where reliance on conventional mild steel persists. Ultimately, effective corrosion management is critical for ensuring safe, sustainable, and resilient infrastructure systems worldwide.

Introduction

Reinforced concrete (RC) has become the backbone of modern infrastructure, combining the compressive strength of concrete with the tensile strength of mild steel reinforcement. Its versatility, cost-effectiveness, and ability to withstand heavy loads have made it the most widely used construction material in buildings, bridges, highways, dams, and marine structures. However, the long-term durability and performance of RC structures are intrinsically linked to the integrity of the embedded steel reinforcement. When mild steel corrodes, the consequences extend beyond the material level, affecting the mechanical, structural, and service performance of the entire system (Wong, et al, 2022).

Corrosion of reinforcement in concrete is primarily driven by environmental factors such as chloride ion penetration, carbonation of the concrete matrix, and exposure to moisture and oxygen. Chloride-induced corrosion is particularly aggressive in coastal and marine environments, where salt-laden air and seawater infiltration initiate localized pitting on the steel surface (Ahmad, et al, 2024). Carbonation, on the other hand, gradually lowers the alkalinity of concrete, breaking down the protective passive film around the reinforcement. Both mechanisms result in the formation of expansive rust products, which exert tensile stresses on the surrounding concrete, leading to micro-cracking, delamination, and eventually spalling of the cover.

The mechanical implications of steel corrosion include a reduction in cross-sectional area, deterioration of bond strength between steel and concrete, and a decline in tensile ductility. These changes compromise the stress transfer mechanism within RC members, leading to premature failure under service and ultimate loads. Structurally, the damage manifests as cracking, stiffness reduction, excessive deflection, and diminished load-bearing capacity, which can endanger both the safety and serviceability of infrastructure. The World Federation of Corrosion (WFC) has

estimated that corrosion-related deterioration of reinforced concrete contributes to a significant portion of the trillions of dollars lost annually to corrosion globally, underlining the socio-economic relevance of the problem (Köhncke et al, 2024).

In recent decades, research has focused on understanding corrosion mechanisms, quantifying their impact on mechanical and structural behavior, and developing prevention and mitigation strategies. Advanced diagnostic techniques, the use of corrosion inhibitors, cathodic protection, epoxy-coated and stainless-steel rebars, as well as alternative materials such as fiber-reinforced polymers (FRP), have been proposed to enhance durability. Nevertheless, in many developing regions, including sub-Saharan Africa, the reliance on conventional mild steel reinforcement persists, often without adequate protection or monitoring systems, making corrosion a pressing engineering challenge (Fuhaid, et al, 2022).

This paper explores the corrosion of mild steel in reinforced concrete structures with an emphasis on its mechanical and structural implications. The study provides a synthesis of existing knowledge, highlights real-world case studies, and examines mitigation techniques that can prolong service life (Lollini, et al, 2021). By bridging the material-level understanding with structural performance considerations, the paper seeks to underscore the importance of corrosion management in achieving sustainable and resilient infrastructure.



Figure 1: Corrosion affected thermo-mechanically treated (TMT) mild steel rod in concrete (Angst, 2023).

Corrosion Mechanisms in Reinforced Concrete (RC) Structures

The corrosion of mild steel reinforcement in reinforced concrete (RC) is an electrochemical process that occurs when the protective environment of concrete is compromised. In its intact state, concrete provides an alkaline environment ($\text{pH} \approx 12.5\text{--}13.5$) due to the presence of calcium hydroxide, which maintains a passive oxide film on the steel surface. This passive film significantly reduces the rate of electrochemical reactions, ensuring long-term durability (Montero, et al, 2023). However, when aggressive agents penetrate the concrete or the pH is reduced, the passivity of steel is destroyed, leading to corrosion initiation.

Chloride-Induced Corrosion

Chloride ingress is the most aggressive corrosion mechanism in RC structures, particularly in marine environments, regions with de-icing salt usage, or industrial zones. Chloride ions penetrate the concrete through pores and micro-cracks, reaching the steel surface. Once the chloride concentration exceeds a threshold level (commonly referred to as the critical chloride concentration), the passive film breaks down locally, initiating pitting corrosion. This form of corrosion is highly localized, producing deep pits on the steel surface that drastically reduce the effective cross-sectional area of reinforcement. Pitting corrosion is especially dangerous because it can remain undetected until advanced damage has occurred, compromising the structural safety of the RC member (Angst, 2023).

Carbonation-Induced Corrosion

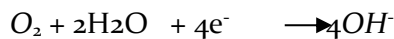
Carbonation is a slow chemical reaction between atmospheric carbon dioxide (CO_2) and calcium hydroxide (Ca(OH)_2) in concrete pore solution, producing calcium carbonate (CaCO_3). This reaction gradually reduces the pH of the pore solution to values below 9, destroying the stability of the passive film on steel. Once depassivation occurs and moisture and oxygen are available, uniform corrosion develops across the steel surface (Ali, et al, 2024). Unlike chloride-induced corrosion, which is localized, carbonation typically causes widespread corrosion, leading to general loss of cross-section and weakening of reinforcement.

Electrochemical Process of Corrosion

The corrosion of steel reinforcement is essentially an electrochemical cell process. At anodic sites on the steel surface, iron dissolves into the pore solution as ferrous ions (Fe^{2+}):



At cathodic sites, the electrons generated are consumed by the reduction of oxygen in the presence of water:



The ferrous ions further react with hydroxyl ions to form ferrous hydroxide, which oxidizes into ferric hydroxide and eventually into hydrated iron oxides (rust). The rust products have 2–6 times the volume of the original steel, generating expansive stresses that crack and spall the surrounding concrete cover (Shevtsov, et al, 2022).

Coupled Action of Moisture, Oxygen, and Temperature

For corrosion to proceed, three key elements must coexist: moisture, oxygen, and an aggressive environment (chlorides or carbonation). Moisture in the form of capillary water or high relative humidity facilitates ionic transport. Oxygen availability at the steel-concrete interface enables cathodic reactions, while elevated temperature accelerates electrochemical kinetics. Seasonal variations and microclimatic conditions therefore significantly influence corrosion rates in RC structures (Shilar, 2024).

Synergistic Effects of Chloride and Carbonation

In many real-world scenarios, chloride ingress and carbonation act simultaneously, compounding the deterioration process. For example, in marine environments, chloride penetration initiates pitting corrosion, while carbonation accelerates the general depassivation of steel, making the reinforcement more susceptible to widespread attack (Wong, 2022). This synergy explains the accelerated deterioration observed in coastal bridges, ports, and offshore platforms.

Manifestations in RC Structures

The visible manifestations of corrosion in RC structures include cracking parallel to reinforcement, delamination of cover concrete, spalling, rust staining, and in advanced stages, exposure of reinforcement bars. At the microscopic level, bond deterioration between steel and concrete weakens the load transfer mechanism (Shalabi, et al, 2024). Over time, these defects compromise the serviceability and safety of RC members, necessitating costly repair or replacement.

Mechanical Implications of Corrosion in RC Structures

The mechanical performance of reinforced concrete (RC) structures is inherently tied to the composite action between concrete and embedded steel reinforcement. Corrosion disrupts this synergy by degrading the mechanical integrity of the steel, weakening the concrete-steel bond, and altering the stress distribution within structural members. The consequences are multifaceted, influencing strength, ductility, stiffness, and ultimately the load-bearing capacity of RC structures (Angst, et al, 2020).

Loss of Cross-Sectional Area of Reinforcement

One of the most direct mechanical effects of corrosion is the reduction of the effective cross-sectional area of reinforcing bars. As rust formation consumes steel, the load-carrying capacity of the reinforcement diminishes. The severity of this reduction depends on the corrosion rate and duration. For example, localized pitting corrosion can remove significant material from critical sections, resulting in stress concentration and premature yielding. Uniform corrosion, though slower, leads to gradual and widespread section loss, reducing overall strength (Li, et al, 2025).

Reduction in Ductility and Toughness

Corrosion alters the microstructure of steel reinforcement, leading to embrittlement. The pits formed during chloride-induced corrosion act as notches, reducing the steel's ductility and fracture toughness. Consequently, corroded bars often exhibit reduced elongation capacity and increased susceptibility to brittle fracture under tensile loads (Leporace-Guimil, et al, 2021). This is particularly critical during seismic events, where reinforcement must undergo large plastic deformations to dissipate energy.

Degradation of Bond Strength

The bond between reinforcement and concrete is essential for composite action. Corrosion weakens this bond in two opposing ways: initially, the expansion of rust products may increase confinement and bond resistance. However, as corrosion progresses, cracking, delamination, and spalling of the cover concrete significantly reduce the bond strength (Chen, et al, 2024). Loss of bond leads to premature slip of reinforcement under loading, impairing structural stiffness and causing serviceability issues such as excessive deflections.

Reduction in Structural Stiffness and Load Capacity

The combined effects of reinforcement area loss, bond degradation, and cover cracking result in reduced flexural and shear stiffness of RC members. Experimental studies have shown that beams with corroded reinforcement exhibit reduced load-deflection capacity, earlier yielding, and diminished ultimate strength compared to uncorroded counterparts. In columns, corrosion-induced degradation can compromise axial load-bearing capacity and stability, increasing the risk of progressive collapse (Fan, et al, 2022).

Influence on Fatigue Performance

Corrosion significantly reduces the fatigue life of reinforcement by introducing surface irregularities that act as stress raisers. Under cyclic loading, such as in bridges and

offshore structures, the initiation and propagation of fatigue cracks are accelerated in corroded bars. Reduced fatigue strength can cause failure even under service loads well below the design capacity, undermining the long-term reliability of RC structures (Angst, et al, 2024).

Effects on Structural Redundancy and Safety

According to American Concrete Institute (ACI) Committee (2019), corrosion alters the overall redundancy and robustness of RC systems. Localized failures in beams, slabs, or joints due to corroded reinforcement can propagate into system-level instability. In critical infrastructure such as bridges, buildings, and retaining walls, these localized mechanical impairments compromise the global safety margin, increasing the probability of catastrophic failure under extreme loads.

Structural Implications of Corrosion in RC Structures

While the mechanical implications of corrosion affect the reinforcement and its interaction with concrete, the structural implications are manifested at the level of the entire reinforced concrete (RC) system (Köhncke, et al, 2024). These consequences extend beyond localized damage to influence durability, serviceability, and the overall safety of structures. Corrosion compromises not only the load-bearing function of individual members but also the continuity, stability, and long-term reliability of structural systems.

Cracking and Spalling of Concrete Cover

The expansive nature of corrosion products exerts tensile stresses on the surrounding concrete. This leads to the initiation of cracks that propagate through the cover, ultimately causing spalling (Montero, et al, 2023). Loss of concrete cover reduces the protective barrier around reinforcement, accelerating further ingress of chloride ions and moisture. Structurally, spalling diminishes confinement, reduces section size, and exposes reinforcement to direct environmental attack, compounding the deterioration process.

Reduction in Load-Bearing Capacity

Structural members such as beams, columns, and slabs rely on the effective interaction of reinforcement and concrete to carry design loads (Ahmad, et al, 2024). Corrosion-induced loss of reinforcement cross-section, combined with bond degradation, leads to significant reductions in load-bearing capacity. In columns, axial load resistance is impaired, whereas in beams and slabs, flexural and shear capacities are diminished. This compromises the ability of RC structures to resist both service and ultimate loads.

Serviceability Issues

Corrosion leads to visible defects such as cracks, rust stains, deflections, and surface delaminations (Ahmad, et al, 2024). These serviceability problems reduce user comfort, increase maintenance costs, and may render structures functionally unfit even before structural collapse occurs. For bridges and highways, this translates to increased repair cycles and reduced operational efficiency. In building structures, corrosion-induced serviceability failures can lead to partial closures, evacuation, and economic disruption (Fan, et al, 2022).

Reduction in Durability and Service Life

The onset and progression of corrosion drastically reduce the expected service life of RC structures. Durability, defined as the ability to withstand environmental actions over time, is compromised as corrosion accelerates deterioration under aggressive exposure conditions such as marine environments or industrial atmospheres (Angst, et al, 2024). Reduced durability directly increases life-cycle costs by necessitating frequent repairs, rehabilitation, or premature replacement.

Structural Instability and Collapse

In severe cases, uncontrolled corrosion can precipitate structural instability and catastrophic collapse. Localized failure of corroded beams or joints may propagate into progressive collapse, especially in statically indeterminate systems with low redundancy (Ali, et al, 2024). Historical failures of bridges, parking garages, and building components illustrate the structural consequences of neglecting corrosion. Such failures not only endanger human lives but also result in significant economic and social losses.

Implications for Structural Safety and Reliability

The reduction in mechanical properties, loss of section, and diminished load capacity collectively reduce the structural safety margin. Corrosion-induced uncertainties in structural performance increase the risk of unexpected failure under normal service loads or extreme events such as earthquakes and wind storms (Ahmad, etb al, 2024). Reliability-based assessments have shown that corroded RC structures exhibit higher probability of failure compared to their uncorroded counterparts, necessitating continuous monitoring and predictive maintenance strategies.

Case Studies of Corrosion in RC Structures

Case studies of reinforced concrete (RC) structures worldwide provide tangible evidence of the severe implications of corrosion. These examples illustrate how material degradation, when left unchecked, progresses from minor defects to major

structural failures. They also emphasize the importance of preventive maintenance, monitoring, and rehabilitation (Fuhaid, et al, 2020).

Highway Bridges

Corrosion in bridges has been widely reported, particularly in regions exposed to de-icing salts and marine environments. The Silver Bridge collapse (1967, Ohio River, USA), though primarily attributed to a fractured eye-bar, highlighted how corrosion-related fatigue can exacerbate localized weaknesses (Ahmad, et al, 2024). Similarly, chloride-induced corrosion in reinforcing steel is a recurring problem in North American highway bridges. Loss of concrete cover, extensive spalling, and reinforcement exposure have forced many bridges into costly repair programs or complete replacement decades earlier than their design service life.

Parking Garages

Multi-storey parking structures often face aggressive exposure due to cycles of wetting and drying, coupled with de-icing salts carried by vehicles (Montero, et al, 2023). The Pipers Row car park in Wolverhampton, UK, experienced extensive deterioration characterized by delamination of slabs and corrosion of reinforcement. Structural assessments revealed significant reduction in flexural capacity, prompting rehabilitation through cathodic protection and concrete repair. These cases underline how corrosion in service environments with high chloride ingress accelerates deterioration of slabs and beams.

Marine Piers and Coastal Structures

RC piers and wharves located in tidal and splash zones are particularly vulnerable to corrosion. The Queensland wharf structures in Australia provide documented evidence of rapid reinforcement corrosion due to chloride attack from seawater (Montero, et al, 2023). Observed damage included cracking of piles, loss of bond, and spalling of cover concrete. Rehabilitation required fiber-reinforced polymer (FRP) wrapping and impressed current cathodic protection systems, demonstrating the high cost of intervention once corrosion becomes widespread (Fuhaid, et al, 2020).

Residential and Commercial Buildings

Residential RC buildings are not immune. In Nigeria and other tropical regions, numerous cases of building distress have been traced to poor-quality concrete, inadequate cover, and exposure to aggressive environments (Montero, et al, 2023). Corrosion-induced cracking of columns and beams has led to partial collapses of buildings in Lagos and Port Harcourt. These failures underscore the role of both material quality and environmental exposure in accelerating corrosion.

Catastrophic Failures

The Champlain Towers South collapse (Fuhaid, et al, 2020) is a recent high-profile example where long-term corrosion and deterioration were identified as major contributors. Investigations revealed extensive reinforcement corrosion, concrete spalling, and compromised slab-column connections. The progressive collapse of this residential tower demonstrates how corrosion, if unaddressed, can escalate into large-scale structural failure with devastating human and economic consequences (Montero, et al, 2023).

Mitigation and Monitoring Strategies

Given the severe mechanical and structural implications of corrosion in reinforced concrete (RC) structures, robust mitigation and monitoring strategies are essential to enhance durability, extend service life, and minimize costly rehabilitation. These strategies can be broadly categorized into preventive measures, protective interventions, and monitoring techniques.

Material-Based Mitigation

High-Performance Concrete (HPC): Increasing the density and reducing the permeability of concrete through the use of supplementary cementitious materials (SCMs) such as fly ash, silica fume, and ground granulated blast furnace slag (GGBFS) significantly slows chloride ingress and carbonation.

- **Corrosion-Resistant Reinforcement:** Stainless steel, epoxy-coated bars, and galvanized reinforcement provide enhanced resistance to corrosion initiation. More recently, fiber-reinforced polymer (FRP) bars have been adopted as non-corroding alternatives in aggressive environments.
- **Adequate Concrete Cover:** Ensuring sufficient thickness of concrete over reinforcement, as stipulated by design codes, delays the arrival of aggressive agents such as chlorides and carbon dioxide at the steel surface.

Structural Protection Techniques

Coatings and Sealers: Surface-applied hydrophobic agents (silane, siloxane) and epoxy coatings reduce ingress of water and chlorides. These methods are particularly effective for bridges and parking garages exposed to de-icing salts.

Cathodic Protection (CP): Both impressed current and sacrificial anode CP systems are effective in halting active corrosion. CP has been widely used in marine piers, bridge decks, and buried RC pipelines.

Electrochemical Treatments: Electrochemical chloride extraction (ECE) and realkalization are applied to restore passivity in reinforcement and reduce chloride concentration in the concrete (Fuhaid, et al, 2020).

Design and Construction Practices

Durability-Oriented Design: Structural design should account for environmental exposure classes, with specifications for water–cement ratio, minimum compressive strength, and cover depth.

- **Quality Control During Construction:** Proper compaction, curing, and adherence to mix design reduce voids and cracks that could otherwise accelerate corrosion.
- **Drainage and Detailing:** Preventing water stagnation through adequate drainage details, slope design, and crack control joints reduces the likelihood of chloride concentration and carbonation.

Monitoring and Assessment Techniques

Effective monitoring is key to early detection of corrosion and timely intervention. Common techniques include:

- i. **Half-Cell Potential Mapping:** Provides a probabilistic assessment of corrosion activity on reinforcement.
- ii. **Linear Polarization Resistance (LPR):** Measures the corrosion current density, allowing quantification of corrosion rate.
- iii. **Concrete Resistivity Measurement:** Assesses the likelihood of corrosion initiation by evaluating the electrical resistivity of concrete.
- iv. **Fiber Optic Sensors and Embedded Corrosion Sensors:** Emerging technologies allow real-time monitoring of structural health, transmitting continuous data on corrosion activity, humidity, and chloride levels.
- v. **Non-Destructive Testing (NDT):** Ground-penetrating radar (GPR), ultrasonic pulse velocity, and infrared thermography detect delamination, cracking, and moisture ingress associated with corrosion.

Maintenance and Rehabilitation Strategies

Preventive Maintenance: Scheduled inspection and minor repairs of cracks, coatings, and sealants extend service life at minimal cost.

- (i) **Targeted Repair and Strengthening:** Techniques such as patch repair, jacketing, and FRP wrapping restore structural integrity after localized corrosion damage.
- (ii) **Lifecycle-Based Asset Management:** Integrating corrosion monitoring data into predictive models enables engineers to optimize repair timing and allocate resources efficiently.

Conclusion

Corrosion of mild steel reinforcement in reinforced concrete (RC) structures is a critical durability challenge that directly threatens structural safety, functionality, and service

life. The underlying mechanisms, particularly chloride ingress and carbonation, progressively undermine the protective environment of concrete and initiate electrochemical reactions that lead to reinforcement deterioration. The consequences manifest as reduced bond strength, diminished ductility, cracking, spalling, and eventual structural instability.

From a mechanical perspective, corrosion reduces the effective cross-sectional area of reinforcement and alters stress distribution within structural elements, thereby compromising their load-bearing capacity. Structurally, it accelerates serviceability failures such as cracking and deflection, while in severe cases it precipitates sudden collapse. Evidence from global case studies highlights the high vulnerability of marine infrastructure, urban high-rise buildings, and bridges in de-icing salt regions, underscoring the socio-economic and safety implications of corrosion.

Although the problem is pervasive, contemporary mitigation strategies such as corrosion-resistant materials, cathodic protection, inhibitors, self-healing concretes, and advanced monitoring technologies provide effective pathways for minimizing risk. With deliberate adoption and integration of these strategies, the durability of RC structures can be greatly enhanced.

Recommendations

To address corrosion challenges in RC structures more effectively, the following recommendations are made:

- (i) **Durability-based design:** Codes of practice should mandate design approaches that account for environmental exposure conditions, minimum cover depth, and the use of supplementary cementitious materials (SCMs).
- (ii) **Use of advanced reinforcement systems:** Adoption of epoxy-coated rebars, stainless steel, galvanized reinforcement, or non-metallic alternatives such as fiber-reinforced polymer (FRP) should be prioritized for aggressive environments.
- (iii) **Regular monitoring and inspection:** Application of non-destructive testing (NDT), half cell potential mapping, and embedded sensors should become standard practice for early detection and intervention.
- (iv) **Preventive maintenance culture:** Asset owners should adopt proactive repair measures including surface treatments, crack sealing, protective coatings, and periodic concrete re-alkalization rather than waiting for extensive damage.
- (v) **Capacity building:** Engineers, contractors, and technicians should receive continuous training in modern corrosion control technologies to improve implementation in practice.

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